

| No. of Trials | Actual Mean Rewards for Each Arm                             | Best Arm | Estimated Mean Reward | Regret                               |
|---------------|--------------------------------------------------------------|----------|-----------------------|--------------------------------------|
| 100           | [ 1.45 1.1 1.98 1.27 -1.43 0.76 -0.67 -2.32 -1.55 -0.59]     | 2        | 155.470658611984      | UCB                                  |
| 500           | [-0.72 -1.73 -1.28 -0.96 -0.44 -0.04 0.73 0.61 -0.3 0.84]    | 9        | 364.16                | ThompsonSampling                     |
| 1000          | [ 0.48 0.94 -0.38 -0.69 1.93 -0.51 -0.75 1.47 -0.23 2.22]    | 9        | 2011.28               | ThompsonSampling                     |
| 1500          | [-0.63 1.76 -1.51 0.89 -1.66 -1.37 -0.82 0.49 -1.54 -0.22]   | 1        | 2329.29               | UCB                                  |
| 2000          | [-0.64 0.97 1.82 0.63 -0.18 -0.8 0.4 -0.53 0.78 1.33]        | 2        | 3333.85               | UCB                                  |
| 2500          | [-0.12 1.14 -0.95 2.01 -0.58 -0.21 0.68 -0.05 -0.55 2.31]    | 9        | 5112.48               | ThompsonSampling<br>And UCB got same |
| 3000          | [-1.51 -0.82 -0.28 -0.47 -0.89 -0.82 -2.23 -0.87 -1.39 1.11] | 9        | 2662.05               | ThompsonSampling                     |
| 3500          | [-1.2 1.13 -0.48 -1.08 -0.21 0.5 -0.64 -0.19 1.59 -0.37]     | 8        | 4949.59               | UCB                                  |
| 4000          | [ 1.25 -0.07 0.83 1.49 -0.4 -1.68 0.23 -0.5 0.35 -0.69]      | 3        | 5357.53               | UCB                                  |
| 4500          | [-0.02 -0.93 0.66 0.97 0.45 -0.06 0.08 -3.21 -1.34 0.52]     | 3        | 3734.15               | UCB                                  |
| 5000          | [ 1.41 -0.91 -1.56 -0.73 1.31 -1.4 -0.23 -2.12 0.98 0.44]    | 0        | 6254.43               | ThompsonSampling                     |